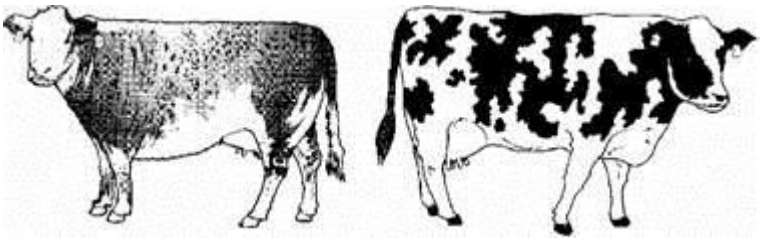
Q1.

Herefords and Friesians are two breeds of cattle.
Herefords produce high quality meat. Friesians produce lots of milk.

The drawings below show a Hereford cow and a Friesian cow.



Hereford cow Friesian cow

(a) (i) The two breeds of cattle are different in appearance from each other.
What causes the variation between the two breeds of cattle?

.....

.....

1 mark

(ii) Suggest **two** environmental factors which can affect the amount of meat or milk cattle produce.

1. .....

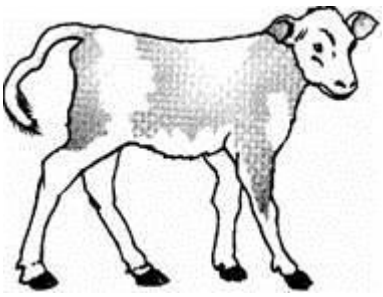
.....

2. .....

.....

2 marks

(b) The drawing shows a calf produced by mating a Hereford bull with a Friesian cow. Cattle bred in this way will produce both high quality meat and a high milk yield.



(i) What term is used to describe this deliberate mating of two different breeds of animals to produce offspring with particular characteristics?

.....

1 mark

(ii) Farmers want their cattle to produce high quality meat and a high milk yield.
Suggest **one other** characteristic which farmers might want their cattle to have.

.....

.....

1 mark
Maximum 5 marks

Q1.

(a) (i) any **one** from

- they have inherited different characteristics
  accept 'selective breeding'
- they have different genes
  *accept 'they have different DNA **or** alleles'*

1

(ii) any **two** from

- amount of food
  accept 'amount of grass'
- type of food
- amount of water
- levels of disease **or** quality of health care
  *accept 'disease' **or** 'health care'*
- levels of stress
- temperature
  accept 'climate'
  *accept 'amount of exercise' **or** 'amount of space'*
  accept 'their hormone levels'

2

(b) (i) selective breeding
*accept 'artificial selection' **or** 'cross breeding'*
*do **not** accept 'inter-breeding'*

1

(ii) any **one** from

- resistance to disease
  *do **not** accept 'good health'*
- fast growth rate
- high fertility
  accept 'fertile'
- long life
- good skin for making leather
- docile temperament
- hardiness **or** resistance to weather

accept 'prize-winning appearance'
accept 'females' ***or*** *'heifers'*
accept 'polled' ***or*** *'no horns'*

1

[5]